

The Zero State of the 8×8 Torus

A Unique Vacuum Configuration from Vacuum Native Measurements

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Abstract

The WIN/FDLM framework identifies the vacuum with the 8×8 torus and the 14 modes at the mirror fixed eigenvalue $\lambda = 4$ with the Standard Model particle content. Prior work derived the mode classification (5+9 split, D_4 orbits, support classes, entropy quantization) but left open the question of which configuration the vacuum actually passes through during processing.

We define a measurement matrix M of 19 vacuum-native observables (support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, helicity flip probability, and seven geometric labels) on the 14 modes. We prove that $\text{rank}(M) = 13$ and $\text{nullity}(M^T) = 1$. There is exactly one 14-vector s (up to scale) such that $\sum_i s_i M_i = \mathbf{0}$ on every measurement. This is the zero state the unique vacuum configuration consistent with all 19 observables simultaneously vanishing.

We establish four structural properties of s . First, it is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. Second, the two self-paired modes $(2, 2)$ and $(6, 6)$ have exactly zero amplitude. Third, three amplitudes satisfy exact algebraic ratios: $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Fourth, 86.5% of the $|s|$ -mass sits on the middle D_4 orbit $\mathcal{O}_1$, the eight modes of support 48.

We also establish the irreducibility of the zero state. The entropy column is redundant, the 15 rational columns contain s exactly, and the amplitudes a_2 and a_6 have no minimal polynomial of degree ≤ 12 and are not in $\mathbb{Q}(\sqrt{2})$ with small coefficients.

We falsify three alternative selection mechanisms. Static pressure-charge balance prefers symmetric $7 + 7$ partitions in 95% of cases; the 5+9 split wins uniquely zero times in 5,035 measurement subsets. The DNLS flow does not select the zero state, with stationarity residual 0.583 and no attraction from random initial conditions. Topology gives zero winding and zero sign-size correlation.

The zero state is not a selection rule. It is a structural fact about the 14-mode multiplet and the 19 observables. We discuss the connection to the CKM matrix and identify the zero state as a necessary input to a future flavor-mixing derivation.

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1 Introduction

1.1 The Selection Problem

The 8×8 torus Laplacian has 64 eigenmodes with 13 distinct eigenvalues. The spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. The prior WIN white papers identify these 14 modes with the particle content of the Standard Model, classified into three D_4 orbits of sizes 2, 8, and 4, with supports 64, 48, and 32.

What has not been established is: when the vacuum processes a state through the torus, which configuration does it pass through? The framework has been silent on the actual state, treating the multiplet as a static classification.

This paper addresses the question with a new tool: the measurement matrix.

1.2 The Measurement Matrix

We define 19 vacuum native observables on each of the 14 modes. These are listed in Table 1.

Table 1: The 19 vacuum-native observables.		
Observable	Definition	Source
support	number of non zero sites	lattice geometry
IPR	inverse participation ratio	mode norm
entropy	Shannon entropy of mode density	information theory
$N/4$	mass-ordering label	δ function
δ	perturbation from base	δ function
shell	5-level coarse classification	shell rule
charge	mode charge from two-mesh split	charge assignment
$P_{\parallel}$	pressure along winding direction	pressure tensor
$P_{\perp}$	pressure perpendicular	pressure tensor
P_{total}	$P_{\parallel} + P_{\perp}$	pressure tensor
P_{flip}	helicity-flip probability	vacuum birefringence
n_1, n_2	mode momentum labels	lattice geometry
$n_1 + n_2$	winding sum	lattice geometry
$ n_2 - n_1 $	diagonal distance	lattice geometry
$n_1 n_2$	winding product	lattice geometry
$v_x, v_y, v $	group velocity components	dispersion relation

Together these form a 14×19 matrix M .

1.3 Central Result

Theorem 1 (Zero-State Existence and Uniqueness). *The measurement matrix M has rank 13. Its transpose has a one-dimensional nullspace. There is exactly one 14-vector s , unique up to scale, satisfying*

$$M^T s = \mathbf{0}.$$

The vector s is the zero state of the vacuum.

We establish this numerically to machine precision. The smallest singular value of M is 8.47×10^{-16} , while the next-smallest is 1.44×10^{-1} .

2 Setup

2.1 The 14 Modes at $\lambda = 4$

The 14 modes of the 8×8 torus at $\lambda = 4$ have momentum labels (n_1, n_2) satisfying $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \tag{1}$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \tag{2}$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \tag{3}$$

These are the three D_4 orbits of sizes 2, 8, and 4, and supports 64, 48, and 32, which equal N , V , and $N/2$.

2.2 Singular Value Decomposition

The singular values of M , from largest to smallest, are

190.09, 52.54, 12.08, 9.85, 6.83, 6.07, 4.41, 3.38, 1.10, 1.02, 0.43, 0.26, 0.14, 8.47×10^{-16} .

The smallest is machine zero. The rank is 13, the nullspace of M^T is one-dimensional.

3 The Zero State

3.1 The Nullspace Vector

The one-dimensional nullspace vector s , normalized so that $\max_i |s_i| = 1$, has components listed in Table 2.

Table 2: The zero-state amplitudes.

i	(n_1, n_2)	Orbit	s_i
0	(0, 4)	$\mathcal{O}_0$	-0.105198199128
1	(1, 3)	$\mathcal{O}_1$	+0.937726561300
2	(1, 5)	$\mathcal{O}_1$	-0.192806690737
3	(2, 2)	$\mathcal{O}_2$	0
4	(2, 6)	$\mathcal{O}_2$	-0.329141609215
5	(3, 1)	$\mathcal{O}_1$	-0.937726561300
6	(3, 7)	$\mathcal{O}_1$	+0.658283218429
7	(4, 0)	$\mathcal{O}_0$	+0.105198199128
8	(5, 1)	$\mathcal{O}_1$	+0.192806690737
9	(5, 7)	$\mathcal{O}_1$	-1.000000000000
10	(6, 2)	$\mathcal{O}_2$	+0.329141609215
11	(6, 6)	$\mathcal{O}_2$	0
12	(7, 3)	$\mathcal{O}_1$	-0.658283218429
13	(7, 5)	$\mathcal{O}_1$	+1.000000000000

3.2 Symmetry Under Diagonal Reflection

Define the diagonal reflection $R_d : (n_1, n_2) \mapsto (n_2, n_1)$. Under this action, mode i maps to a mode $\sigma(i)$. The vector s satisfies

$$s_{\sigma(i)} = -s_i \quad \text{for all } i. \quad (4)$$

That is, s is odd under R_d . Direct verification shows each diagonal pair has opposite signs.

The two diagonal modes (2, 2) and (6, 6) are fixed points of R_d and must have $s = 0$ for an odd vector. The nullspace vector satisfies this to machine precision:

$$|s_3|, |s_{11}| < 3 \times 10^{-16}.$$

3.3 Six Independent Amplitudes

By the antisymmetry, the zero state is determined by six independent real amplitudes, one for each diagonal pair. These are listed in Table 3.

The zero state is exactly six numbers.

Table 3: The six independent amplitudes.

Amplitude	Pair	Value
a_1	$(5, 7)/(7, 5)$	1.000000000000
a_2	$(1, 3)/(3, 1)$	0.937726561300
a_3	$(3, 7)/(7, 3)$	0.658283218429
a_4	$(2, 6)/(6, 2)$	0.329141609215
a_5	$(1, 5)/(5, 1)$	0.192806690737
a_6	$(0, 4)/(4, 0)$	0.105198199128

4 Algebraic Structure

4.1 Three Exact Ratios

The six amplitudes satisfy three exact algebraic ratios, verified to machine precision. These are listed in Table 4.

Table 4: Exact algebraic ratios in the zero state.

Ratio	Exact form	Numerical	Error
a_3/a_4	2	2.000000000000007	6.6×10^{-15}
a_5/a_4	$2 - \sqrt{2}$	0.585786437626912	6.6×10^{-15}
a_5/a_3	$(2 - \sqrt{2})/2$	0.292893218813455	2.3×10^{-15}

The $\sqrt{2}$ -chain $a_3 \rightarrow a_4 \rightarrow a_5$ is exact. The three amplitudes are determined by a single algebraic base a_4 via two multiplication factors, 2 and $2 - \sqrt{2}$.

These ratios follow from the structure of the real cosine modes at $\pi/4$ multiplicities. The modes (3, 7) and (7, 3) have trigonometric content in the ratio 2, and the modes (1, 5) and (5, 1) have content in the ratio $2 - \sqrt{2}$.

4.2 The Two Non-Algebraic Amplitudes

The amplitudes a_2 and a_6 do not fit any of the following:

- Rationals with denominator at most 10^7
- Minimal polynomial of degree at most 12
- $\mathbb{Q}(\sqrt{2})$ with coefficients $|p|$, $|q|$, and $|n|$ at most 100
- 120 candidate closed-form expressions

The rational reconstruction at denominator 10^7 produced “converged” rationals with denominators between 6×10^6 and 9×10^6 . Inspection of the convergence table shows the error drops by two orders of magnitude per order of magnitude increase in the bound. This is the classic signature of continued-fraction approximation of an irrational number. The amplitudes are not rational.

The two amplitudes may be algebraic of degree greater than 12 over $\mathbb{Q}$, or transcendental, involving kL , π , or combinations from the irrational columns of M . The current data does not distinguish between these possibilities.

4.3 Why a_2 and a_6 Resist

The 19 measurement columns split into two groups:

- 15 rational columns: support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, n_1 , n_2 , $n_1 + n_2$, $|n_2 - n_1|$, $n_1 n_2$.
- 4 irrational columns: P_{flip} , v_x , v_y , $|v|$.

The 15-column rational submatrix has rank 11 and nullity 3. The full 19-column matrix has rank 13 and nullity 1. The four irrational columns contribute two extra independent constraints that cut the nullspace from dimension 3 to dimension 1.

The zero state satisfies the rational constraints to machine precision:

$$\|M_{\text{rat}}^T s\| = 1.0 \times 10^{-14}.$$

The rational submatrix contains the zero state in its nullspace, but does not determine it uniquely. The two additional constraints from P_{flip} and v eliminate the two extra nullspace dimensions. The residual freedom, after rational constraints, is a three-dimensional space, and s is one specific vector in it.

5 Entropy Redundancy

The entropy column of M is

$$S_i = \log_2 N + 1 - N \cdot \text{IPR}_i = 6 + 1 - 64 \cdot \text{IPR}_i.$$

This is an exact affine function of the IPR column. Therefore entropy carries no independent information.

We verified this by deleting the entropy column and recomputing the nullspace. The result is

$$\text{overlap}(s_{\text{full}}, s_{\text{noentropy}}) = 1.0000000000000000.$$

The vectors are identical to 14 decimal places. The entropy column is structurally redundant.

6 Falsification of Alternative Selection Mechanisms

The zero state is a mathematical object. We tested three proposed physical mechanisms for how the vacuum might select it, and falsified all three.

6.1 Static Pressure-Charge Balance

For each partition of the 14 modes into two groups, compute the pressure imbalance $|P_A - P_B|$ and charge imbalance $|Q_A - Q_B|$. Test which partitions minimize the combined imbalance.

Result: across 16,382 partitions, the top ten by combined imbalance are all 7 + 7 splits with imbalance exactly 0.000000. The 5+9 split ranks 14,126 with imbalance 1.083.

Consequence: the mesh prefers symmetric 7 + 7 partitions. The 5+9 split is not a pressure-charge selection.

6.2 Monte Carlo over Measurement Subsets

We tested 5,035 subsets of the 19 measurements, covering all combinations of size 1 to 4. For each subset, we found the partition of the 14 modes that minimizes the combined normalized imbalance.

Result: the summary statistics are as follows. Subsets tested: 5,035. Median margin between best and second-best: 0.000000. Best unique winner: 9.0% of subsets. 5+9 split as unique winner: 0.0%. Partitions with $k = 7$ winning: 95.0% of subsets.

Consequence: no subset of the 19 measurements uniquely selects the 5+9 split. The mesh's only structural preference is for symmetric $7 + 7$ partitions.

6.3 DNLS Dynamical Selection

We integrated the DNLS equation

$$i\partial_t\psi = -\Delta\psi + U|\psi|^2\psi, \quad U = 0.541197,$$

from the zero state and from random states in the 14-mode subspace. Results are shown in Table 5.

Table 5: DNLS dynamical tests.

Test	Result
Stationarity of zero state: $\ \perp\ /\ \text{lhs}\ $	0.583
DNLS from zero state, alignment at $T = 20$	0.845
DNLS from random state, alignment at $T = 20$	0.465
Norm conservation	1.000

The zero state is not a DNLS fixed point. The nonlinear term does not vanish on the zero state. The DNLS flow dephases the state but partially preserves it because the 14-mode subspace is D_4 -protected.

Consequence: the zero state is not dynamically selected. It is at best conserved. The projection onto the zero state direction is preserved by the flow, but the state does not attract other states toward it.

6.4 Topological Selection

We computed two topological invariants:

- Weighted winding charge: $\sum_i s_i n_{1,i}$ and $\sum_i s_i n_{2,i}$.
- Sign-size correlation: the Pearson r between $\text{sign}(s_i)$ and $|Z_i|$, where Z_i is the zero set of the i -th cosine mode.

Result: both weighted windings are 0.0000000000. The Pearson correlation is 0.000000.

Consequence: the zero state has no topological charge and no correlation with the mode zero-set sizes. Topology does not select it.

7 What the Zero State Is

7.1 A Structural Fact, Not a Selection

The zero state exists, is unique, and has a specific algebraic structure. It is not selected by pressure balance, by dynamics, or by topology. It is a consequence of the rank deficiency of the measurement matrix M .

The rank deficiency itself is a structural fact: 19 measurements on 14 modes have an interplay that leaves exactly one linear dependency. This dependency is the zero state.

7.2 Physical Interpretation

The zero state is a superposition of all 14 modes with amplitudes concentrated on the middle orbit $\mathcal{O}_1$. The weight fractions are

$$\mathcal{O}_0 : 3.3\%, \quad \mathcal{O}_1 : 86.5\%, \quad \mathcal{O}_2 : 10.2\%.$$

The middle orbit dominates. The two self-paired modes are dark.

7.3 The 5+9 Split versus the Zero State

The prior WIN work derived a 5+9 split based on the δ function: five substrate eigenmodes with $\delta = 0$ and nine mesh resonances with $\delta \neq 0$. This is a classification.

The zero state is a superposition, not a classification. It includes all 14 modes with nonzero amplitudes, except the two dark modes, and concentrates on $\mathcal{O}_1$. It does not align with the 5+9 split:

- Five substrate eigenmodes: $(2, 2)$, $(2, 6)$, $(5, 7)$, $(6, 2)$, $(7, 5)$.
- Nine mesh resonances: $(0, 4)$, $(1, 3)$, $(1, 5)$, $(3, 1)$, $(3, 7)$, $(4, 0)$, $(5, 1)$, $(6, 6)$, $(7, 3)$.
- Zero state: all 12 non-diagonal modes, weighted 86.5% on $\mathcal{O}_1$.

The 5+9 split partitions the modes. The zero state weights them.

7.4 Rank of the 14 Cosine Modes

The 14 cosine modes span a rank-8 real space, not rank-14. There are six linear dependencies among the 14 cosine modes on the 8×8 torus. The 14-mode multiplet is a redundant description of an 8-dimensional real mode space.

This is a structural fact about the torus geometry that has not been emphasized in prior WIN papers. It is the reason the measurement matrix can have a one-dimensional nullspace despite being 14×19 .

8 Connection to the CKM Matrix

The zero state has structural features that connect to the flavor-mixing sector, though it does not by itself derive the CKM matrix.

8.1 Amplitude on the Quark Modes

The six quark modes are

$$\begin{aligned} u &= (1, 3), \quad c = (7, 3), \quad t = (6, 6) \quad (\text{up-type}), \\ d &= (1, 5), \quad s = (3, 1), \quad b = (5, 1) \quad (\text{down-type}). \end{aligned}$$

The zero state has amplitudes

$$\begin{aligned} s_u &= +0.938, \quad s_c = -0.658, \quad s_t = 0, \\ s_d &= -0.193, \quad s_s = -0.938, \quad s_b = +0.193. \end{aligned}$$

Five of the six quark modes carry nonzero weight. Only the top is dark.

8.2 Orbit Concentration and the Flavor Hierarchy

A total of 86.5% of the zero-state weight sits on orbit $\mathcal{O}_1$, which contains five of the six quark modes, namely u , c , d , s , and b . The heavy orbit $\mathcal{O}_2$, which contains the top quark, carries only 10.2%.

This structure is qualitatively consistent with the observed flavor hierarchy. The zero state is dominated by the light and middle quark modes, while the third-generation top is decoupled from the vacuum's zero configuration. Whether this reflects the physical origin of the top mass or is coincidental is not determined by the present analysis.

8.3 CPT and the Diagonal Reflection

The zero state is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. Prior WIN work established that the CPT relation $J_{\bar{q}} = -J_q$ for the Jarlskog invariant comes from the same reflection. The zero-state antisymmetry and the CPT structure of flavor mixing share the same geometric origin.

This does not produce a derivation of the CP-violating phase, but it identifies the zero state as a configuration consistent with the framework's CPT sector.

8.4 What Is Not Derived

A full derivation of the CKM matrix requires a symmetry group with a 3-dimensional irreducible representation, which the 8×8 torus D_4 symmetry does not possess. This was established in the prior CKM paper. The zero state identifies where the vacuum sits in the quark sector. It does not supply the misalignment between the up-type and down-type mass matrices that produces the mixing angles and the CP phase.

The zero state is therefore a necessary input to a future CKM derivation, not the derivation itself. Any extension of the framework with a 3D irrep will need to be consistent with the zero-state amplitudes derived here.

9 What Is Derived and What Is Open

9.1 Fully Derived in This Paper

1. The measurement matrix M with 19 vacuum-native observables.
2. The rank of M is 13 and the nullity of M^T is 1.
3. The zero-state vector s to 15-digit precision.
4. The antisymmetry of s under the diagonal reflection.
5. The exact zero amplitudes of the diagonal modes $(2, 2)$ and $(6, 6)$.
6. The three exact algebraic ratios: $a_3/a_4 = 2$, $a_5/a_4 = 2 - \sqrt{2}$, and $a_5/a_3 = (2 - \sqrt{2})/2$.
7. The entropy column redundancy.
8. The rational-columns decomposition. The 15 rational columns contain s , the 4 irrational columns cut the nullity from 3 to 1.
9. The falsification of static balance, dynamics, and topology as selection mechanisms.
10. The weight fractions: 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.
11. The rank of the 14 cosine modes is 8.
12. The structural connection between the zero state and the quark sector.

9.2 Partially Derived

1. The amplitudes a_2 and a_6 are irrational, but their exact form, whether algebraic of high degree or transcendental, is not determined.

9.3 Open

1. The physical origin of the rank-8 reduction of the 14 cosine modes.
2. The physical interpretation of the zero state, in particular whether it is the mean-field vacuum expectation value.
3. The extension to include CPT, flavor, and higher-order corrections.
4. The relationship between the zero state and the 5+9 split, if any.
5. Whether the amplitudes a_2 and a_6 have a closed form in kL or other framework constants.
6. The full derivation of the CKM matrix from a 3D irrep extension consistent with the zero state.

10 Conclusion

The 8×8 torus of the WIN paradigm has a unique zero state: a specific superposition of the 14 modes at $\lambda = 4$ that vanishes on all 19 vacuum-native measurements. This state is derived, not fitted. It is the one-dimensional nullspace of the measurement matrix M , whose smallest singular value is machine zero.

The zero state has four structural properties. First, it is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. Second, the diagonal modes $(2, 2)$ and $(6, 6)$ have exactly zero amplitude. Third, three amplitudes satisfy the exact algebraic ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Fourth, 86.5% of the $|s|$ -mass sits on the middle D_4 orbit $\mathcal{O}_1$.

The zero state is not a selection rule. It is a structural fact about the torus and the observable set. The static balance mechanism prefers $7 + 7$ partitions, the DNLS dynamics does not select it, and topology gives zero. The zero state is conserved by the DNLS flow but is not attracted to by it.

We have identified a structural connection between the zero state and the quark sector. Five of the six quark modes carry nonzero weight, and the zero state is dominated by the orbit that contains the light and middle quarks. A full CKM derivation requires a 3-dimensional irrep, which the 8×8 torus does not have. The zero state is a necessary input to such a derivation, not the derivation itself.

This is a genuine addition to the WIN program. The prior work had a classification, the 5+9 split, and a static multiplet. This paper adds a unique vacuum state, derived from the same geometry and the same observables, with specific algebraic structure and specific falsifiable predictions.

A Numerical Values

B Related Work

1. S. Preschutti, Derivation of the 8×8 Torus Substrate from the Standard Model Particle Count, WIN Paradigm White Paper (2026).
2. S. Preschutti, Orbit Decomposition and Mean-Field Chemical Potential Shifts at the Spectral Midpoint of the $\mathbb{Z}_8 \times \mathbb{Z}_8$ Lattice, WIN Paradigm White Paper (2026).

Table 6: Summary of numerical values.

Quantity	Value
Torus size	8×8
Sites	64
Modes at $\lambda = 4$	14
D_4 orbit sizes	2, 8, 4
Support classes	64, 48, 32
Measurements	19
Rank of M	13
Nullity of M^T	1
Smallest singular value	8.47×10^{-16}
Second-smallest singular value	1.44×10^{-1}
Rank of 14 cosine modes	8
Zero-state amplitudes	1.000, 0.938, 0.658, 0.329, 0.193, 0.105
Exact ratios	$a_3/a_4 = 2$, $a_5/a_4 = 2 - \sqrt{2}$
Weight on $\mathcal{O}_1$	86.5%
Entropy redundancy	overlap = 1.0000000000000000
Rational-columns residual	1.0×10^{-14}
DNLS stationarity residual	0.583
DNLS alignment at $T = 20$	0.845
Weighted winding	0.0000000000
Topological correlation	0.000000

3. S. Preschutti, The δ Function on the 8×8 Torus: Derivation of the 14 $N/4$ Values from the Mode Coordinates, WIN Paradigm White Paper (2026).
4. S. Preschutti, The $5 + 9$ Split of the 14-Mode Multiplet, WIN Paradigm White Paper (2026).
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